

QUESTIONS FOR INTRODUCTION TO MACROECONOMICS EXAM – June 2019

INTRODUCTORY PART (2 QUESTIONS)

Q1

Government interventions in the economy can be summarized under three distinct functions. Which of the following propositions **does not** refer to a government function?

- (a) The allocation function, i.e. the government offers public goods and services.
- (b) The stabilization function, i.e. the government, via monetary or fiscal policy, guarantees a global equilibrium.
- (c) The allocation function, i.e. the government redistributes the society's richness based on some criteria.
- (d) The stabilization function, i.e. the government intervenes in markets while paying attention not to increase inflation and unemployment and to achieve a balance of payments at equilibrium.

Answer (c)

Q2

Which of the following propositions is correct?

- (a) Macroeconomics does not necessarily take into consideration all legal transactions made in the economy
- (b) John Maynard Keynes is considered as the founding father of macroeconomics.
- (c) Macroeconomics is nothing else than the aggregation of microeconomic behaviors.
- (d) The Great Depression of the 1930s has shown that the markets are able to adjust themselves and no Government intervention is necessary.

Answer (b)

PART 1 – (8 QUESTIONS)

Q3

Assume a world with only two countries, A and B. We have the following information:

Country A		Country B	
GDP at market prices	110	Capital depreciation	15
Government expenditure	25	Subsidies	10
Gross investment	20	Indirect taxes	5
Net exports	15	National income	80
Factor income paid to foreigners	10		
Factor income received from foreigners	15		
Indirect taxes	10		

We also know that in Country A, capital depreciation is equal to subsidies minus indirect taxes. Which of the following propositions is correct?

- a) Country A's NNP at factor prices equals 115 and country B's GDP at market prices

- equals 100.
- b) Country A's national income equals 110 and country B's GDP at factor prices equals 120.
 - c) Country A's GNP at market prices equals 105 and country B's GDP at market prices equals 95.
 - d) Country A's NNP at factor prices equals 115 and country B's GDP at market prices equals 95.

Answer (d)

Q4

Which of the following propositions concerning GDP is correct?

- (a) GDP is a good measure of a country's income especially in periods of strong economic integration.
- (b) GDP per capita is a strongly correlated indicator to a handful of other variables influencing life quality.
- (c) GDP per capita allows to measure income inequalities in a country.
- (d) GDP is the value of production at market prices of every good and service produced by nationals of a country during a specific period of time.

Answer (b)

Q5

Assume that in 2010 an employee earns 50000 euros. In 2011 her nominal salary increases by 4000 euros. Taking 2010 as the base year, the CPI in 2011 is 108. By how much did the real salary of the employee increase?

- (a) 2%.
- (b) 3%.
- (c) 0%.
- (d) 1%.

Answer (c)

Q6

Suppose there is a price index based on the annual expenditure of a representative consumer (or representative consumption basket) in Switzerland. The following table details the expenditure structure for year 2012 (base year) and 2018.

Good	Quantity in 2012 (in thousands)	Quantity in 2018 (in thousands)	Unit price in 2012 (CHF)	Unit price in 2018 (CHF)
Sushi	1	2	5	30
Clothes	1	2	20	20
Books	2	4	10	10
Coffee	2	5	10	10

Commented [FS1]: In order to make this question less evident, I suggest you add a second column of quantities in 2018 and you put some of the wrong answers evaluated on the grounds of this column. OK

Mobile subscription	3	3	50	100
Heating	4	4	5	20

The price index for Switzerland in 2018 is :

- (a) 470.
- (b) $470/235 \times 100 = 200$.
- (c) $570/235 \times 100 \approx 242$.
- (d) 570.

Answer (b)

Q7

Consider a standard representative production function with capital and labor as inputs, with constant returns to scale and diminishing marginal returns. If the capital decreases by 15% following a natural disaster, what can you infer?

- (a) The amount of labor input increases.
- (b) The production decreases by 15%.
- (c) The total factor productivity decreases.
- (d) The production per capita decreases by less than 15%.

Answer (d)

Q8

Consider the following countries, France, UK and India with an average GDP growth rate of 0.5%, 1% and 7% per year respectively. Under these conditions, how many years will it take for each country to double its GDP (approximately)?

- (a) France takes 140 years, UK takes 70 years and India takes 10.
- (b) France takes 14 years, UK takes 7 years and India takes 1 year.
- (c) France takes 1.4 years, UK takes 0.7 years, India takes 0.1 years.
- (d) The GDP of France cannot double as its average growth rate is less than 1%, UK takes 70 years, India takes 14 years.

Answer (a)

Q9

Which of the following propositions regarding minimum wages is correct?

- a. A minimum wage is a form of price ceiling.
- b. The introduction of a minimum wage would always lead to a fall in the demand for labour by firms.
- c. A minimum wage would not lead to increased unemployment if it is set low enough.
- d. In order to avoid an increase in unemployment, instead of imposing a minimum wage, the government can promote higher labour participation, which would

encourage firms to set higher wages.

Answer (c)

Q10

Assume that the representative production function is given by $Y(L) = 30L - 0.01L^2$. The labor supply function is given by $L^S = 100w$ where w is the real wage. The labor force is 1500. Which of the following propositions is correct?

- (a) The unemployment rate is 25%.
- (b) The inactive population is higher than the employed population.
- (c) A minimum wage of 8 would have no consequence in this set up.
- (d) One employee earns less than its reservation wage.

Answer (c)

SECOND PART – (6 QUESTIONS)

Q11

Suppose the amount of economic uncertainty diminishes, so that households now have a reduced incentive to save. Which will be the impact on the interest rate and on the amount of loanable funds at the new equilibrium?

- a. The interest rate increases, the amount of loanable funds increases.
- b. The interest rate increases, the amount of loanable funds decreases.
- c. There will be no impact neither on the interest rate nor on the amount of loanable funds as public savings will compensate private ones.
- d. The impact on the interest rate is uncertain, while the amount of loanable funds increases.

Answer (b)

Q12

Assume in France an increase in the price of housing is observed. Which of the following propositions is correct?

- a. Existing homeowners are worse-off and will not be able to repay their mortgage.
- b. If financial markets are efficient, this might simply reflect house scarcity.
- c. Banks will have an incentive to reduce lending to would-be homeowners.
- d. A period of declining house prices will necessarily follow.

Answer (b)

Q13

Which of the following propositions regarding inflation and deflation is correct?

- a. Borrowers benefit from deflation, as the value of their debt decreases in real terms.

Commented [FS2]: I am a bit lost with this question. Besides, I am not sure we gave this kind of information to our students. Either we erase it or we modify it into something like:

Assume in France an increase in the price of housing is observed. Which of the following propositions is correct?

- (a) Existing homeowners are worse-off and will not be able to repay their mortgage.
- (b) If financial markets are efficient, this might simply reflect house scarcity.
- (c) Banks will have an incentive to reduce lending to would-be homeowners.
- (d) A period of declining house prices will necessarily follow. OK

- b. Inflation transfers wealth from lenders to borrowers.
- c. Falling prices benefit consumers and are therefore always good for the economy.
- d. Inflation makes it difficult for consumers and firms to attain the message about scarcity of resources (sent by relative prices), and is therefore always bad for the economy.

Answer (b)

Q14

The Central bank increases the monetary supply by 4%. We suppose that the velocity of money is constant over time. According to the quantitative theory of money, which scenario is **not** plausible?

- (a) Nominal GDP increases by 4%.
- (b) Inflation increases by 4%.
- (c) Nominal GDP increases by 3% and inflation by 1%.
- (d) Real GDP increases by 0% and inflation by 4%.

Answer (c)

Q15

Consider that the Central Bank purchases 4 billion worth of government bonds. If reserve requirement is $r = 20\%$, the maximum variation in money supply will be equal to:

- (a) -20 billions.
- (b) -8 billions.
- (c) 20 billions.
- (d) We cannot answer this question because we don't have any information regarding the interest rate.

Answer (c)

Q16

Which of the following propositions is correct?

- (a) Scriptural money is a type of impalpable money that can be transformed at all time into commodity money; it is all the assets one holds in a bank.
- (b) Monetary policy is the set of measures the Central Bank undertakes in order to influence the demand of money and to supervise the efficiency of the monetary system.
- (c) The Central Bank's main goal is to promote price stability in order to fight unemployment.
- (d) None of the above.

Answer (d)

THIRD PART – (8 QUESTIONS)

Q17

In an open economy, GDP equals 50, investment equals 20 and consumption equals 10. Furthermore, net capital outflow in this economy is positive. Based on the given information, it can be stated that:

- (a) Government expenditure as well as national savings both equal 20.
- (b) Government expenditure is greater than 20 and national savings are lower than 20.
- (c) Government expenditure and national savings are both lower than 20.
- (d) Government expenditure is smaller than 20 and national savings are greater than 20.

Answer (d)

Q18

On January 29th 2016 you find the following information: CHF/USD: 1.023, that is 1CHF = 1.023 USD. On April 5th 2016, you are informed that CHF/USD: 0.9585. From the point of view of Switzerland, which of the following propositions is correct?

- I Quotes are presented in indirect quotation.
- II Quotes are presented in direct quotation.
- III There was an appreciation of the Swiss Franc during the period from 29/01 to 5/04.
- IV There was a depreciation of the Swiss Franc during the period from 29/01 to 5/04.

- (a) Only statements I and III are correct.
- (b) Only statements I and IV are correct.
- (c) Only statements II and III are correct.
- (d) Only statements II and IV are correct.

Answer (b)

Q19

Suppose you have news from two friends, Paul and Helen, who are currently interns abroad. Paul, who is in India tells you that the famous IKEA lamp costs 360 Rupees and that the exchange rate is 1 CHF = 70 INR. Helen, who is in Hong Kong bought the same lamp for 32 Hong Kong Dollars (HKD). The exchange rate is 1 CHF= 8 HKD. In Geneva, the same lamp costs 4 Swiss Francs (CHF). Given the available information, if the law of one price is verified in the medium run, you can state that (with respect to the Swiss Franc):

- (a) The Rupee will depreciate and the Hong Kong Dollar will appreciate.
- (b) The Rupee will appreciate and the Hong Kong Dollar will depreciate.
- (c) Only the Rupee will appreciate.
- (d) Only the Rupee will depreciate.

Answer (d)

Q20

Take the following statistics:

	Price of a Big-Mac in local currency	Nominal exchange rate on the currency market
USA	USD 3	-
Japan	Yen 270	100 Yen = 1 USD
China	Yuan 12	10 Yuan = 1 USD
Switzerland	CHF 6	1 CHF = 1 USD

Commented [FS3]: Again, we need simpler figures so that calculations are easier. Please check whether 100Yen = 1USD, 10Yuan = 1USD and 1CHF = 1USD are still fine. Yes, they are OK

According to the Big Mac index, which of the following is the correct case?

- (a) One can expect an appreciation of all three currencies with respect to the dollar.
- (b) The Swiss franc is overvalued with respect to the dollar.
- (c) The Yen and the Yuan are both overvalued with respect to the dollar.
- (d) One can expect an appreciation of the dollar with respect to the Yuan.

Answer (b)

Q21

Let's consider an open economy which is initially at equilibrium. Which of the following government policies would lead to an increase of both the production and price level in the short run?

- (a) A tax decrease for the households of this economy.
- (b) A decrease in military spending.
- (c) A decrease of the public deficit.
- (d) A decrease of the money supply.

Answer (a)

Q22

If the government decreases its budget deficit, which of the following propositions is correct?

- (a) Net exports increase and there is a real depreciation.
- (b) Net exports increase and there is a real appreciation.
- (c) Net exports decrease and there is a real depreciation.
- (d) Net exports decrease and there is a real appreciation.

Answer (a)

Q23

In a small open economy, suppose that the government introduces an investment tax which leads to an increase in public savings. Keeping everything else constant, what would be the expected effect of this measure?

- (a) The interest rate decreases and there is a real depreciation of the domestic currency.
- (b) The effect on the interest rate is uncertain and the net capital outflow increases.

- (c) The interest rate decreases and the net exports decrease.
- (d) Net capital outflow is not affected.

Answer (a)

FOURTH PART – (8 QUESTIONS)

Q24

Which of the following propositions regarding the Keynesian multiplier is correct?

- a. If two countries were identical except for the share of low-income households, then the country with the higher share would have a smaller multiplier.
- b. The multiplier is constant over the business cycle.
- c. An increase in exports leads to a higher multiplier.
- d. Taxation and imports are “leakages” from the circular flow of income, which reduce the size of the multiplier.

Answer (d)

Q25

Which of the following propositions regarding fiscal policy is correct?

- a. Expansionary fiscal policy (e.g. increasing the government deficit or reducing the surplus) always has a stabilising effect on the economy.
- b. Unemployment benefits and taxes automatically increase government spending and cut taxation in a downturn, while they trim spending and raise taxes in a boom. These are therefore automatic stabilisers.
- c. In a recession, the aim of a government fiscal expansion is to over-ride the effects of automatic stabilisers.
- d. As a family worried about mounting debts should cut spending and save more, so should an economy adopt austerity measures when its debt level is high, to restore its public finances to balance.

Answer (b)

Q26

Consider a scenario where the Bank of England views the UK economy to be overheating and is attempting to slow the economy down using monetary policy. Which of the following statements regarding the effects of an interest rate rise is correct?

- (a) It leads to higher bond prices, which results in higher demand for UK bonds.
- (b) It leads to higher demand for GBP, which results in an appreciation of the GBP.
- (c) It leads to the UK exports becoming cheaper and imports becoming more expensive.
- (d) It has opposing effects on the UK's aggregate demand (AD): it discourages investment, which lowers AD, but results in cheaper imports, which boosts AD.

Answer (b)

Q27

Right before the outbreak of the 2008 crisis, the Federal Reserve:

- (a) Was worried about the potentially negative effects of the lack of regulation and introduced some capital controls on the US capital market.
- (b) Was worried about an increase in unemployment and reduced the interest rate.
- (c) Was worried about a predictable decline in aggregate demand and increased money supply.
- (d) Was worried about inflationary pressures and reduced money supply.

Answer (d)

Q28

Consider an economy initially at the long run equilibrium. According to the Phillips curve/relation, what can you expect if during the second period the inflation is lower than expected?

- (a) In the short run, the unemployment rate increases. In the long run, the unemployment rate goes back to its initial level. The expected inflation decreases.
- (b) In the short run, the unemployment rate decreases. In the long run, the unemployment rate goes back to its initial level. The expected inflation increases.
- (c) In the short run, the unemployment rate decreases. In the long run, the unemployment rate goes back to its initial level. The expected inflation remains the same.
- (d) In the short run, the unemployment rate increases. In the long run, the unemployment rate stays at this higher level. The expected inflation increases.

Answer (a)

Q29

What justifies the importance of international coordination?

- (a) The first government to implement an expansionary budgetary policy grows faster than the next. The start-up of policies must thus be synchronized.
- (b) If fiscal stimulus policies aren't synchronized, economic stability will be lost.
- (c) It allows to decide which domestic firms will benefit from subsidies in order for every country's GDP to grow at the same rate.
- (d) The best situation is met when each government implements an expansionary fiscal policy. However, each government's best interest is to not cooperate.

Answer (d)

Q30

What does a liquidity trap imply?

- (a) Economic agents experience a rise in their saving rate.

- (b) Any attempt by the central bank to boost the economy via monetary policies will be fruitless.
- (c) An optimistic atmosphere encourages economic agents to use their savings.
- (d) The demand for money is perfectly inelastic.

Answer (b)